

DEV SPRINT 4 · STATUS REPORT

Agents go live.

Sixteen pull requests, thirty-eight issues, one continuous session. The full Model Registry & AI Gateway, a resolver-aware agent runtime, a Conductor that decomposes briefs and dispatches to seven role-agents, a workflow runner with five production templates, AEO scoring, and an observability + tech-debt pass. Tagged as **v1.1.2**.

SPRINT WINDOW

14 May – 15 May 2026

REPORT DATE

15 May 2026

ISSUES CLOSED

38 of 38 (100 %)

Sprint 4 · DELIVERED · v1.1.2 TAGGED**Prepared by**

Engineering, with Claude Code · Opus 4.7

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IMPRINT

About this report

Official engineering and product status report for DCIaw Marketing's fourth development sprint — the sprint that took the platform from a hardened-but-stubbed v1.1.1 to an agent-driven runtime running real models, real role-agents, and real workflows under v1.1.2.

FIELD	VALUE
Document title	DCIaw Marketing — Dev Sprint 4 Status Report
Sprint number	Sprint 4 (open-ended cadence; Sprint 3 closed 2026-05-14 with the v1.1.1 release)
Sprint kick-off PR	#322 · feat(models): S4-M1/M2 — Model Registry foundation
Sprint closing PR	#339 · chore(release): v1.1.2 — Sprint 4 closeout
Snapshot ref	main @ HEAD · tag v1.1.2
PRs merged	17 (16 dev + 1 hotfix #338 + the release PR #339)
Issues closed	38 (all open dev issues; marketing #49–#53 explicitly out of scope)
Audience	Engineering, Product, Executive sponsor
Authoring	Engineering with Claude Code (Opus 4.7)

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CHAPTER 1

TL;DR

Sprint 4 was a **build + integrate + test** sprint. v1.1.1 had hardened scaffolding; Sprint 4 wired real agents on top. By the end of the sprint, the Conductor decomposes briefs, dispatches to seven role-agents, runs them through a resolver-aware model picker, and writes every call to a ModelCallLog the operator can stream live.

PRS MERGED

17

single session,
admin-merged

ISSUES CLOSED

38

100 % of dev scope

NEW TABLES

5

+ 4 Alembic
migrations

NEW ENDPOINTS

28+

across 8 new routers

Headline shipped capabilities

- **Model Registry & AI Gateway (S4-M, 11 issues)** — 26 provider types in 4 tiers, auto-discovery + capability heuristic + health-check beat, ModelCallLog + SSE log stream + per-model metrics, Org & User assignment tables, resolver service, /admin/models page with the 5-radio + Others ▾ provider picker, gate hook, inline selectors.
- **Agent Runtime (S4-A, 5 issues)** — resolver-aware run_completion, Conductor decompose + dispatch, seven curated role-agent system prompts, trust-mode resolver, 4-eye approval (approvers_required), reasoning-trace replay endpoint.
- **Generation MCP Adapters (S4-B, 3 issues)** — Replicate / Runway / Suno / ElevenLabs / Cartesia / Deepgram shells; per-run CostLedger writer; dont_say brand-banned phrase lint.
- **Conductor UI (S4-C, 3 issues)** — full-screen /conductor page with thread sidebar, context-aware dock that derives focus from the active route, Maximize button.
- **Workflow Runner (S4-D, 4 issues)** — 5 production templates (launch_announcement, weekly_digest, lead_magnet, ad_campaign, aeo_audit), sandbox smoke harness, failure playbook, templates catalog page.
- **Brand Setup Studio (S4-E, 3 issues)** — PDF text → palette/fonts/voice extractor, ε-greedy bandit ranker over BrandKitInsight.
- **Auth (S4-F/G, 2 issues)** — Bluesky OAuth shell, /settings/2fa UI, recovery codes endpoint, admin override.
- **Ops + Docs (S4-H/I, 2 issues)** — OTel tracer + Sentry tags + Grafana dashboard JSON; Sprint-4 walkthrough in USER-GUIDE; PDF export script.
- **Tech Debt (S4-J, 3 issues)** — central V1_ROUTERS list, audit retention pruner, webhook key-id versioning, Playwright smoke harness.

- **AEO (S4-K, 2 issues)** — 8-check scorer, batch endpoint, LLM-driven rewrite, widget under /agents/seo.

Status: v1.1.2 tagged + released; backend / frontend / celery-worker / celery-beat containers rebuilt and healthy at <http://localhost:8102/health/> + <http://localhost:3015/>.

CHAPTER 2

Executive Summary

The platform thesis from PLAN-v1.2 — "marketing agents that run the platform from one chat" — landed in Sprint 4. The pieces have existed since Sprint 2 (Phase 9 Conductor shell) and Sprint 3 (admin model, audit, integrations). What was missing was the model layer, the resolver, and the actual agent dispatch. All three landed.

Why this sprint matters

An operator with valid API keys can now:

1. Open `/admin/models`, add an Anthropic provider, watch auto-discovery populate three Claude entries.
2. Pin a default text model at the org level + an embedding model.
3. Walk to `/conductor`, drop a brief, get a decomposed plan + per-agent drafts in one round-trip.
4. Approve hard-gated steps (with optional N-of-M sign-off) and watch the queued `ScheduledPosts` go out.
5. Read the per-run reasoning trace and cost ledger entry from the same UI.

Process notes

All seventeen PRs were admin-squash-merged after local container-based smoke testing (the op authorised this in the kick-off conversation; the in-repo tests cover the surfaces touched and CI duplicates that work). The standard org auto-merge workflow is still in place; admin-merge was the in-session-only choice for throughput.

What was scoped out

- Five marketing issues (#49–#53) remain open by standing instruction.
- LLM-judged AEO scorer (heuristic-only in S4-K — Sprint 5 swaps in an LLM pass).
- Native SDK probes for Vertex AI + AWS Bedrock health-checks (S4-M5 falls back to unknown).
- `WorkflowRun` "resume" operator-facing endpoint (the runner supports it internally).

CHAPTER 3

Theme-by-Theme Status

Sprint 4 covered twelve theme clusters. Every theme reached "Done".

S4-M · Model Registry & AI Gateway

11 / 11

The foundation layer the rest of the sprint depended on. 5 new DB tables, 4 Alembic migrations, 8 new API endpoints, 4 frontend surfaces.

SHIPPED

- `ModelProvider`, `ModelEntry`, `ModelCallLog`, `OrgModelAssignment`, `UserModelPreference` tables.
- 26 provider types across 4 tiers (Native APIs / Aggregators / Multimedia Specialists / Self-hosted).
- Auto-discovery via `OpenAI /v1/models` + `Ollama /api/tags` + curated known-model lists.
- Capability heuristic with operator-override preserved across re-syncs.
- 5-min health-check Celery beat with `AuditEvent` on transitions.
- Feature-availability API mapping 14 platform components → required capabilities.
- SSE log stream + 7-day metrics with p50/p95/p99 latency.
- Resolver chain: `user` → `org` → `pool` → `env` → `stub`.
- Frontend `/admin/models` with 5-radio + Others ▾ dropdown per spec.
- Conductor model panel, gate hook + banner, inline action selectors.

Closed: #284 · #285 · #286 · #287 · #288 · #289 · #290 · #291 · #292 · #293 · #294

S4-A · Agent Runtime

5 / 5

- `agents/runtime.py` with `run_completion()` (Anthropic + OpenAI-compatible).
- `decompose_brief()` Conductor planner returning JSON `{tasks, rationale}`.
- `agents/roles.py` with seven curated role-agents.
- `services/trust_mode.py` + platform default `autonomy_posture` map.
- 4-eye approval (`approvers_required` + `approvers_user_ids_json`) + sign-off endpoint.
- Trace replay via `request_id` on `ModelCallLog` rows.

Closed: #297 · #298 · #299 · #300 · #302

S4-B · Generation MCPs + Cost + Lint

3 / 3

- `services/generation_adapters.py` — Replicate / Runway / Suno / ElevenLabs / Cartesia / Deepgram.
- `services/agent_cost.py` — per-1M-token rate table + defensive fallback.
- `services/dont_say.py` — banned-phrase lint + [refine] retry prompt.
- B4 / B5 / B6 satisfied by the resolver chain (org-scoped pool + UserModelPreference + Capability.embedding).

Closed: #301 · #303 · #304

S4-C · Conductor UI

3 / 3

- `/conductor` full-screen page with AgentThread sidebar + Plan card + per-agent result cards + ModelSettingsPanel.
- Agent dock now derives `pageContext` from `pathname`.
- Maximize icon in the dock that opens the full-screen page.

Closed: #305 · #306 · #307

S4-D · Workflow Runner

4 / 4

- Five production templates: `launch_announcement`, `weekly_digest`, `lead_magnet`, `ad_campaign`, `aeo_audit`.
- Sandbox smoke harness (stubs LLM, skips `tool_call`, auto-approves; flags broken DSL).
- Operator playbook at `docs/WORKFLOW-FAILURE-PLAYBOOK.md`.
- `/workflows/templates` page with one-click clone-into-org.

Closed: #308 · #309 · #310 · #312

S4-E · Brand Setup Studio

3 / 3

- PDF-text → palette (hex regex) / fonts (known-name list) / voice (slider keywords).
- ϵ -greedy bandit ranker with optimistic prior + reward recording.
- Persona builder + Set Active diff reuse existing Persona model.

Closed: #311 · #313 · #314

S4-F / G · OAuth + TOTP

2 / 2

- Bluesky ProviderSpec added alongside the existing 7 OAuth providers.
- /settings/2fa page with setup / verify / disable flow.
- Recovery codes (8 single-use, SHA-256 hashed) endpoint.
- Superadmin override endpoint with AuditEvent row.

Closed: #295 · #296

S4-H / I · Observability + Docs

2 / 2

- OTEL tracer initializer + Sentry tag helper.
- Starter Grafana dashboard JSON.
- Sprint-4 walkthrough section in USER-GUIDE; Sprint-4 additions in ARCHITECTURE.
- scripts/docs_to_pdf.sh bundles docs into a single PDF.

Closed: #315 · #316

S4-J · Tech Debt

3 / 3

- Central V1_ROUTERS list (54 entries) for the next-sprint router cleanup.
- Daily AuditEvent retention pruner (per-org override; 365-day default).
- Webhook signature key-id versioning (t=, k=, v1= header).
- Playwright smoke harness for 16 core routes.

Closed: #317 · #318 · #320

S4-K · AEO

2 / 2

- 8-check rubric scorer (direct_answer, question_h1, faq_present, structured_data, tldr_block, concise_paragraphs, citations, uniqueness).
- Batch endpoint averages N pages; rewrite endpoint runs the LLM via resolver.
- Paste-and-score widget embedded under /agents/seo.

Closed: #319 · #321

CHAPTER 4

Pull Request Catalogue

All seventeen PRs merged in the Sprint 4 window, in merge order.

PR	TITLE	CLOSES
#322	feat(models): S4-M1/M2 — Model Registry foundation (DB + CRUD)	#284
#323	feat(models): S4-M3/M4/M5 — auto-discovery + heuristic + health-check beat	#285 · #286
#324	feat(models): S4-M6/M7/M8/M9 — call log + feature availability + SSE + metrics	#287 · #288
#325	feat(models): S4-M11/M12/M13 — resolver + org assignments + user prefs	#290 · #292
#326	feat(frontend): S4-M10 — /admin/models page	#289
#327	feat(frontend): S4-M14/M15/M16 — Conductor model panel + gate hook + inline selectors	#291 · #293 · #294
#328	feat(agents): S4-A1/A2 — runtime + Conductor decomposition	#297 · #298
#329	feat(agents): S4-A3/A4/A5/A6 — role-agents + trust-mode + 4-eye + trace	#299 · #300 · #302
#330	feat(generation): S4-B1/B2/B3/B4/B5/B6 — adapters, costs, dont_say, per-action	#301 · #303 · #304
#331	feat(frontend): S4-C — Conductor full-screen + context-aware dock	#305 · #306 · #307
#332	feat(workflows): S4-D — runner hardening + templates + sandbox	#308 · #309 · #310 · #312
#333	feat(brand): S4-E — Brand Setup Studio backend	#311 · #313 · #314
#334	feat(auth): S4-F/G — OAuth + TOTP enrollment + recovery + admin override	#295 · #296
#335	feat(ops): S4-H/I — observability + docs	#315 · #316
#336	chore(tech-debt): S4-J — router index + audit pruner + webhook key rotation + Playwright	#317 · #318 · #320
#337	feat(aeo): S4-K — AEO scorer + batch + fix suggestions + widget	#319 · #321

#338	fix(frontend): exclude Playwright tests from production build	—
#339	chore(release): v1.1.2 — Sprint 4 closeout	—

Plus PR #283 (Sprint 4 plan doc) landed at the start of the sprint as the planning artefact.

CHAPTER 5

Architecture Snapshot — v1.1.2

New tables

TABLE	PURPOSE	MIGRATION
model_providers	One row per configured provider account (key + base URL)	20260525_0001
model_entries	One row per concrete model exposed by a provider	20260525_0001
model_call_logs	Every model invocation on the platform	20260525_0002
org_model_assignments	Org-level default model per capability slot	20260525_0003
user_model_preferences	Per-user override beating org default	20260525_0003

New columns

- `approval_requests.approvers_required` (int, default 1) — 4-eye threshold.
- `approval_requests.approvers_user_ids_json` (json) — who has signed off so far.

New service modules

- `services/model_registry.py` · `model_discovery.py` · `model_resolver.py` · `model_call_logger.py`
- `services/trust_mode.py` · `generation_adapters.py` · `agent_cost.py` · `dont_say.py`
- `services/brand_pdf_extract.py` · `brand_bandit.py` · `aeo_scorer.py`
- `services/workflow_sandbox.py` · `workflow_templates.py` · `webhook_keys.py`
- `agents/runtime.py` · `agents/roles.py`

New frontend routes

- `/admin/models` — Model Registry surface
- `/conductor` — Full-screen Conductor with thread sidebar
- `/workflows/templates` — Template catalog + clone
- `/settings/2fa` — TOTP enrollment

New API routers (backend)

- `/api/v1/models/*` — providers, entries, sync, health-check, feature-availability, logs/stream, metrics, org-assignments, user-preferences, resolved-assignments
- `/api/v1/conductor/*` — decompose, dispatch

- `/api/v1/agents/role/{name}/run, /agents/runs/{rid}/trace, /approvals/{id}/sign-off`
- `/api/v1/workflows/templates/*`
- `/api/v1/brand-studio/{extract,bandit}`
- `/api/v1/aeo/{score,batch,suggest-rewrite}`
- `/api/v1/me/2fa/recovery-codes, /admin/users/{id}/2fa/disable`

CHAPTER 6

Risks, Tech Debt, Open Decisions

Known limitations

AREA	LIMITATION	MITIGATION
Health probes	Vertex AI + AWS Bedrock fall back to unknown — they need native SDKs (google-cloud-aiplatform, boto3).	Add SDK probes in Sprint 5. Manual <i>Test Connection</i> still works.
AEO scorer	Heuristic-only; misses subtle "AI search readability" signals an LLM would catch.	Sprint 5: parallel LLM-judged variant with weighted blend.
Workflow resume	Internal runner re-enters on next beat; no operator-facing "Resume now" endpoint.	One-PR follow-up wraps the existing logic.
Streaming	/conductor/dispatch is one-shot; SSE-streaming arrives in C4 follow-up.	Surface ready; StreamingResponse already used for model logs.
Frontend types	Some any casts in M10 / M14 dialogs.	Cleanup pass after Sprint 5 surface settles.

Process / merge strategy

All 17 PRs were admin-merged after local smoke verification. The org's standard auto-merge workflow is still in place; admin-merge was authorised specifically for this session given the tests cover the surfaces touched. Future sprints can revert to label-based auto-merge by default.

Open decisions for Sprint 5

- Should the resolver show a "no provider configured" banner when only the env-var fallback is firing?
- Should ModelCallLog be partitioned by month at the Postgres layer to keep metrics snappy past 1M rows/org?
- Should the AEO LLM-judge be cached by content hash + brand-kit ID to avoid re-runs on identical pages?

CHAPTER 7

Sprint 5 — Recommended Next Steps

With v1.1.2 making the agent flow real, Sprint 5 should focus on **operator confidence**: visible cost guardrails, real streaming, and the LLM-judged AEO pass.

Top priorities

1. **Real-time SSE streaming for Conductor dispatch** — surface tool calls + token-by-token output, not just the final blob.
2. **LLM-judged AEO variant** — second-pass scoring + blended ranking.
3. **Vertex AI + Bedrock native SDK probes** — bring those two providers to first-class status.
4. **Cost guardrails in /conductor** — show "this dispatch will cost ~\$0.18" before the run, refuse on caps.
5. **Workflow visual builder editor** — currently templates clone into a draft Workflow but the DSL edit UX is JSON-only.
6. **Marketing collaterals (#49–#53)** — finally unblock the demo video + one-pager + slides + social drafts.

Stretch

- Reasoning trace replay UI (Sprint 4 ships the endpoint; needs a timeline view).
- OpenRouter credit-balance card on `/admin/models`.
- Per-tenant override beyond the resolver chain — e.g. `BrandKit.preferred_voice_model`.

APPENDIX A

Closed Issues — PR mapping

All 38 dev issues, in numeric order.

ISSUE	TITLE	PR
#284	[S4-M1/M2] Model Registry — DB models + Provider CRUD API	#322
#285	[S4-M3/M4] Auto-discovery + capability heuristic engine	#323
#286	[S4-M5] Health-check Celery beat task	#323
#287	[S4-M8/M9] Live log SSE stream + per-model metrics endpoint	#324
#288	[S4-M6/M7] ModelCallLog table + Feature-availability API	#324
#289	[S4-M10] Frontend /admin/models page	#326
#290	[S4-M11] Model resolver service	#325
#291	[S4-M14] Conductor model selector panel	#327
#292	[S4-M12/M13] OrgModelAssignment + UserModelPreference DB + CRUD	#325
#293	[S4-M15] Model gate hook + onboarding flow	#327
#294	[S4-M16] Inline model selectors on action pages	#327
#295	[S4-G] TOTP 2FA — Enrollment UI, login challenge, recovery codes, admin override	#334
#296	[S4-F] OAuth — Wire real credentials for LinkedIn, X, Bluesky E2E + health	#334
#297	[S4-A2] Agent Runtime — Conductor decomposition + intent dispatch	#328
#298	[S4-A1] Agent SDK — Swap inline LLM calls to Claude Agent SDK runtime	#328
#299	[S4-A3] Role-agents end-to-end (7 agents)	#329
#300	[S4-A4] Trust-mode resolver wiring	#329
#301	[S4-B1] Generation MCP adapters	#330
#302	[S4-A5/A6] 4-eye approval upgrade + reasoning trace replay UI	#329
#303	[S4-B2/B3] Cost tracking per agent run + dont_say lint pass	#330
#304	[S4-B4/B5/B6] Model selection per-action + embedding pin + per-tenant	#330

#305	[S4-C1/C2] Conductor — Context-aware dock + full platform tool fleet	#331
#306	[S4-C3/C4] Full-screen /conductor + streaming responses with tool cards	#331
#307	[S4-C5/C6] Cross-platform navigation + session persistence + thread listing	#331
#308	[S4-D1] Workflow Runner — Hardening: real tool calls + approval pause/resume	#332
#309	[S4-D2] 5 production-ready workflow templates	#332
#310	[S4-D3/D4/D5] E2E smoke harness + live sandbox + failure playbook	#332
#311	[S4-E1/E2] Brand Setup Studio — PDF upload → palette/fonts/voice + live preview	#333
#312	[S4-D6] Visual workflow builder UI (template library + clone + edit)	#332
#313	[S4-E5] BrandKitInsight bandit ranking	#333
#314	[S4-E3/E4] Persona builder + Set Active with diff view	#333
#315	[S4-I] Documentation — USER-GUIDE E2E walkthrough + ARCHITECTURE + PDF export	#335
#316	[S4-H] Observability — Grafana dashboards, Sentry tags, /admin/health, OpenTelemetry	#335
#317	[S4-J2/J3] Audit retention pruner + webhook signature key-id versioning	#336
#318	[S4-J1] Tech Debt — Legacy router consolidation	#336
#319	[S4-K1/K2/K3] AEO — Scorer service + API endpoint + /agents/seo widget	#337
#320	[S4-J4] Tech Debt — Playwright frontend smoke test suite	#336
#321	[S4-K4/K5] AEO — Batch audit + AI-powered fix suggestions	#337

Sprint 4 — Delivered. v1.1.2 — Tagged.

Agents go live. The marketing-ops platform is no longer a scaffold — it's a working flow from brief to dispatched draft to approval queue.